Navdeep Dhull

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Professional Experience:

Stealth AI Startup- *Sr. Gen AI Engineer*, **May 2025 – Present**

- Reduced LLM inference latency by 45% daily active users by implementing vLLM for continuous batching and deploying quantized Llama-3 models on AWS EC2, after identifying token bottleneck issues during peak traffic spikes
- Improved RAG retrieval precision by 28% by designing a hybrid search pipeline combining dense embeddings (OpenAI text-embedding-3) with sparse BM25 keyword scoring and a cross-encoder reranker
- Saved $3,000 monthly in API expenditure by implementing semantic caching with Redis and routing low-complexity user queries to smaller, fine-tuned open-source models
- Automated compliance tracking for financial documents by building a multi-agent framework using LangGraph, allowing agents to extract, verify, and cross-reference structured tables from unstructured

One Four Seven- *Machine Learning Engineer,* **Nov 2020 – Dec 2023**

- Improved click-through rate (CTR) by 14% by training and deploying a BERT-based text classification model for personalized content recommendations, addressing cold-start recommendation challenges for new users
- Decreased model deployment cycle time from 2 weeks to 48 hours by establishing automated CI/CD pipelines using MLflow and Docker for model packaging, taking ownership of the company's first MLOps framework
- Processed 5M+ daily event logs efficiently by optimizing PySpark ETL data pipelines on AWS EMR, resolving memory bottleneck errors during peak batch processing windows
- Enhanced named entity recognition (NER) accuracy by 19% by fine-tuning RoBERTa on domain-specific customer support tickets, replacing a legacy regex-based extraction system

Projects:

Enterprise Multi-Agent Code Migration and Documentation Assistant

- Built a multi-agent framework where specialized LLM roles (Code Analyzer, Refactoring Agent, Unit Test Generator, Documentation Writer) communicate via a directed cyclic graph to execute complex, multi-step software engineering tasks autonomously
- Resolved context loss and cyclic looping when handling large legacy repositories exceeding standard token context windows. Owned the design of state persistence and human-in-the-loop review nodes
- Automated the translation of legacy Python 2 codebase to Python 3 for a 500,000-line enterprise repository, reducing manual migration effort by 65% and achieving a 94% test-pass rate on generated unit tests
- **Tech Stack:** LangGraph, Python, Llama-3-70B, ChromaDB, FastAPI, Docker

Low-Latency Multimodal RAG and Semantic Caching Engine

- Architected a retrieval-augmented generation engine utilizing dense vector search combined with sparse BM25 indexing, coupled with an embedding-based semantic cache layer to short-circuit redundant user queries
- Scaled system performance to handle 1,000 concurrent requests while cutting median response latency down to 320ms, and reducing overall third-party LLM operational overhead by 40%
- **Tech Stack:** vLLM, OpenAI GPT-4o, Pinecone, Redis, Hugging Face Cross-Encoders, AWS ECS

Skills:

Generative AI: LangChain, LlamaIndex, OpenAI API, Anthropic Claude, Hugging Face Transformers, PEFT (LoRA/QLoRA), vLLM, Ollama || **Architecture and MLOps:** RAG Systems, Vector Databases (Pinecone, Milvus, Chroma, pgvector), MLflow, Docker, Kubernetes, CI/CD pipelines || **Backend Engineering:** Python, PyTorch, TensorFlow, FastAPI, SQL, Apache Spark, Kafka || **Cloud Infrastructure:** AWS (SageMaker, Lambda, S3)

Education:

MS in Business Analytics and Artificial Intelligence from The University of Texas at Dallas **Dec 2025**
Bachelor's in Business Administration from Guru Gobind Singh Indraprastha University **Nov 2020**